THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY
Department of Electronic and Computer Engineering
ELEC 1100

Laboratory 5: Sensors & MCU (5%)

A) Objectives:

- To get familiar with the Arduino IDE coding environment.
- To construct the Nano-Board circuit.

B) Hardware and Software:

- Line (bright sensing) sensors, DC motors.
- LM7805 voltage regulator, L293 H-bridge motor driver, 74HC14 inverter.
- Nano-Board (Arduino compatible) and open-source Arduino Software (IDE).

C) Experimental Procedures:

Experiment 1: Building the Robot Car bottom part (~30min)

Step 1: Download the **"Robot Car Guide_Bottom Assembly"** file from your Canvas lab page. A demonstration video for the bottom assembly is also available on Canvas for your reference.

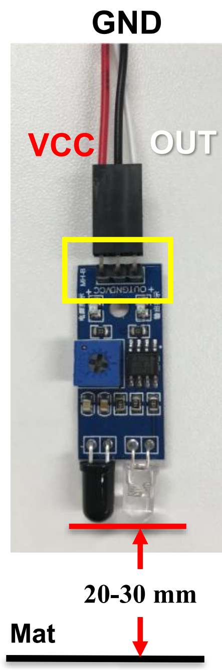
Step 2: Assemble the bottom part of your Robot Car following the guide or demo video. **Do not attach the plastic wheels yet**—adding them now will make it difficult to fit the car body back into the project box.

****** TA Check 1: Show the bottom part of your Robot Car to your TA ******

Experiment 2: Line sensor characteristics (~10 mins)

The line sensor consists of an IR emitter and a photo-diode. It works by illuminating IR on a surface from the emitter, the photo-diode picks up the reflected IR and, based on its intensity, determines the reflectivity of the surface. This allows the sensor to detect a dark line on a pale surface or a pale line on a dark surface.

Moreover, this is a digital sensor, meaning that its output (OUT: white wire) voltage is either "high" (5V) or "low" (0V).



Warning: Do not mix up the wire connections.
Wrong connections will damage the line sensor.

Step 1: Take one sensor and a 3-wire cable, and make the connection as shown in the photo (as in the yellow square, VCC should connect to the red wire while GND to the black wire).

Step 2: Plug the pin header at the other end of the cable into your breadboard wherever available. Each pin shall occupy one individual row.

Step 3: Use a DC power supply to generate 5V. Connect the "+" terminal to **VCC** and the "–" terminal to the **GND** of your sensor.

Step 4: Connect the sensor "OUT" to CH1 of DSO.

You have a dark mat with white lines. Position the line sensor **20-30 mm** above the mat and move it across the white line and dark surface. The sensor should differentiate between them: a **dark surface** will output **5V** (high), while a **white surface** will output **0V** (low).

Step 5: If the sensor cannot differentiate between the dark and white surfaces, use the screwdriver from the project box to adjust the **blue** variable resistor on the line sensor until it can.

****** TA Check 2: Show a demo to your TA that your sensor can differentiate the white line from the dark surface.*****

Experiment 3: Make your Nano-Board blink (~10 min)

In this experiment, you will start to use the Arduino Software (IDE) to write programs and upload them to your board.

You are given a Nano-Board, an IC socket, and a USB cable as below:

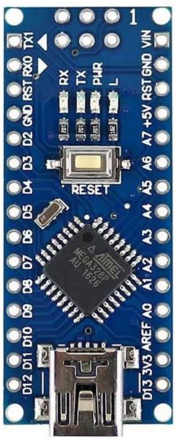
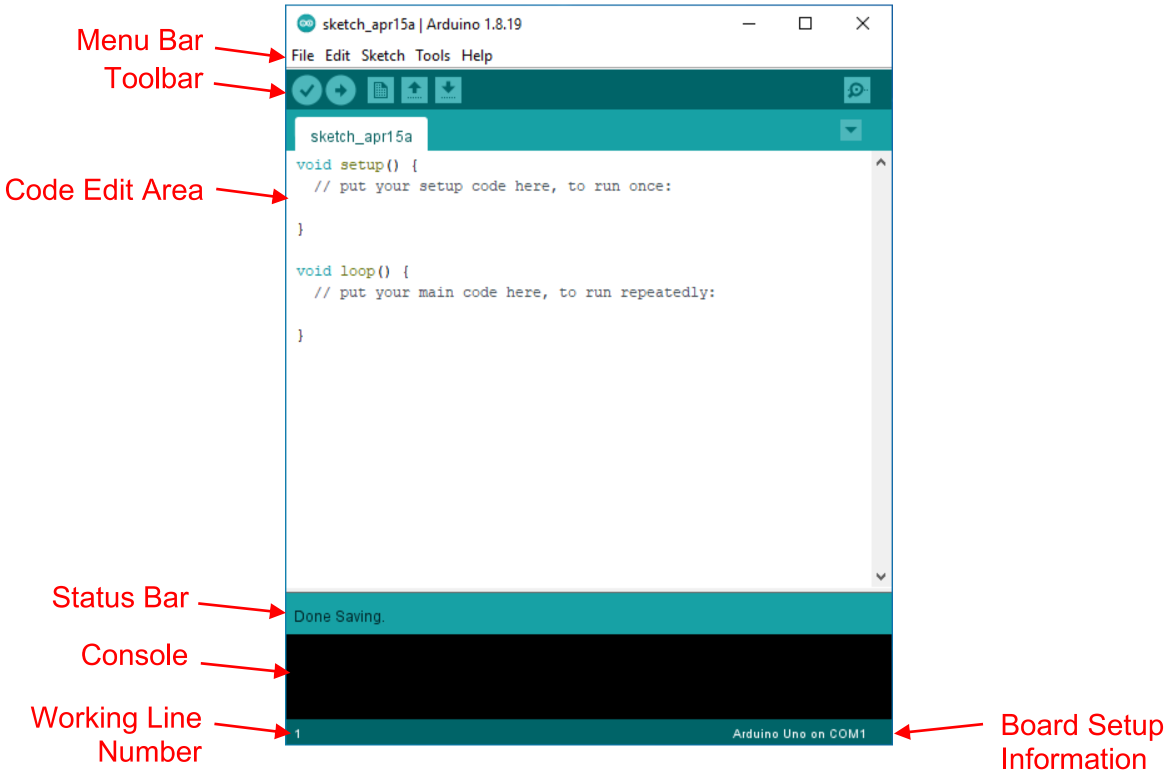


Figure 1. Arduino-compatible Nano-Board

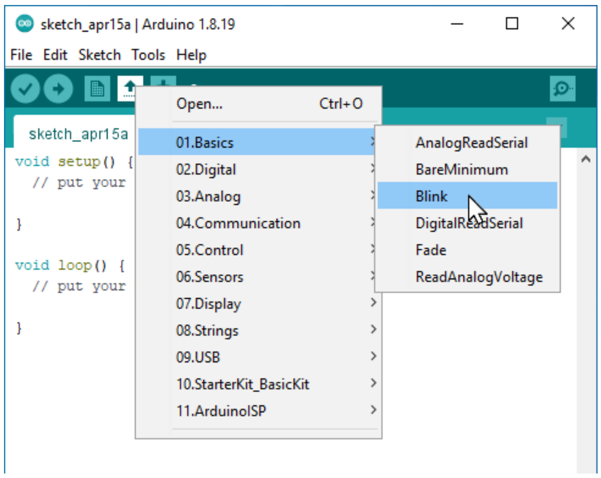
Figure 2. IC Socket

Figure 3. Mini-B to Type-A USB cable

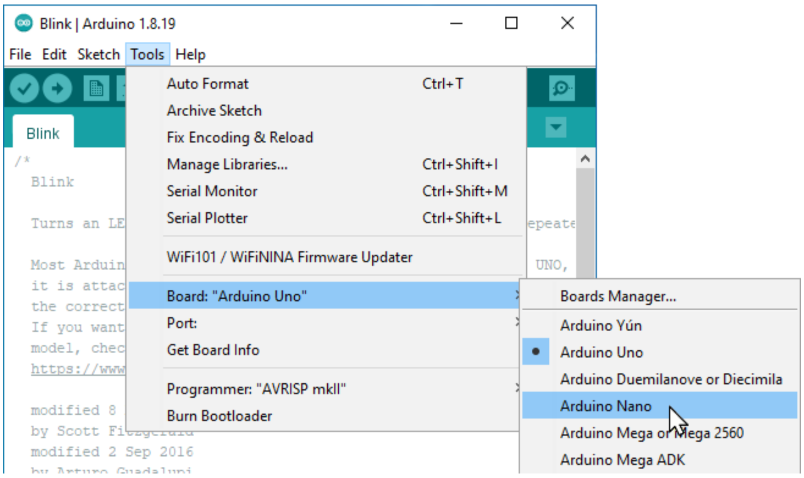
Step 1: Run the Arduino IDE on the desktop of your lab computer. If this is the first time you run the Arduino IDE, you should see a tab (sketch) filled with the two basic Arduino functions: the *setup()* and *loop()*.



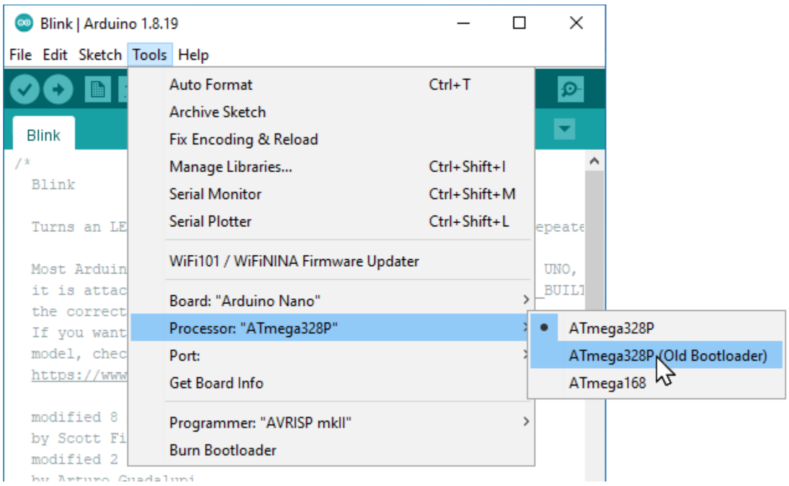
Step 2: Open the LED blink example sketch: **> 01.Basics > Blink**. You should see a new window with the Blink program.



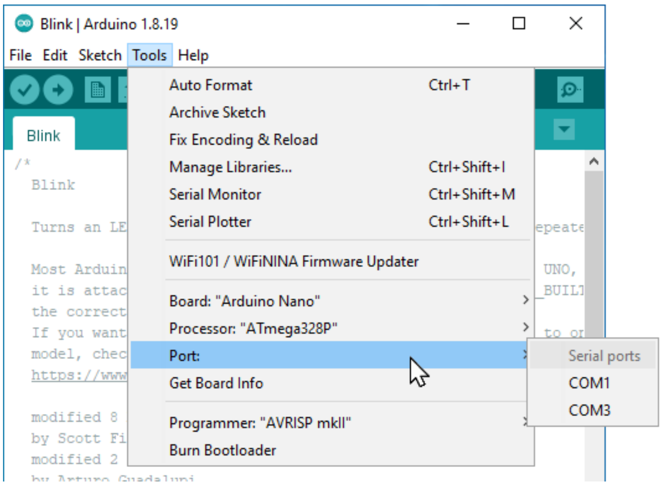
Step 3: Select the board: **Tools > Board > Arduino Nano**.



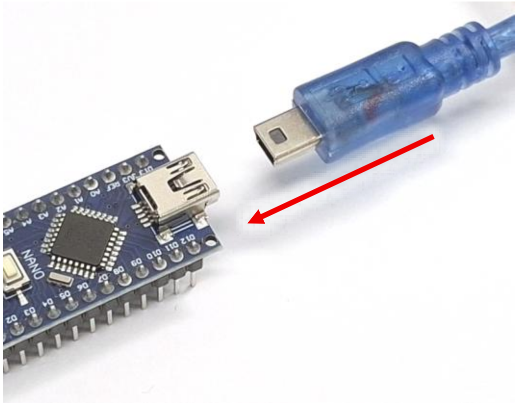
Step 4: Select the processor: **Tools > Processor > ATmega328P (Old Bootloader)**.

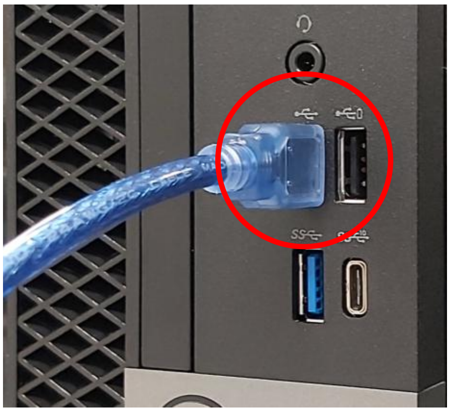


Step 5: Check and note the existing ports available in the PC: **Tools > Port**.

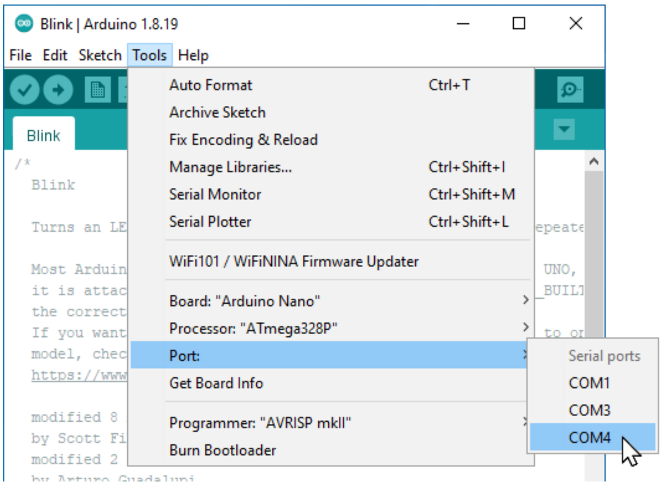


Step 6: Now, connect your Nano-Board to a computer's USB port (use the top one as shown) via provided USB cable. *For first-time connection, it may take a longer time for Windows to establish communication with the device.*





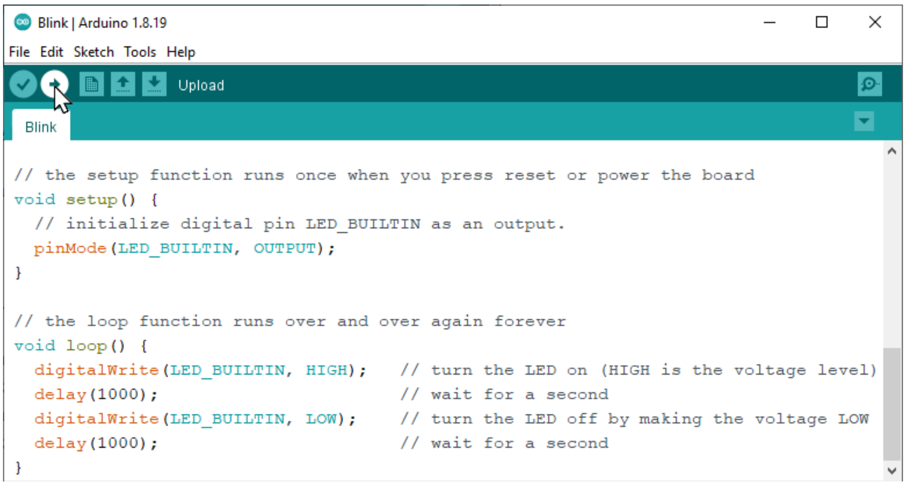
Step 7: Check again the Port list and select the newly added COM port. The new port number is computer-dependent. *If you don't have the new port, seek TA for help.*



Step 8: Click the **Upload**  button to upload the code to the Nano-Board. Wait a few seconds until the message "Done uploading." appears in the status bar.

If there is no response, **disconnect** the USB cable from the Arduino board. Re-select the processor by navigating to: **Tools > Processor > ATmega328P**. Connect the USB cable to the Arduino board, then upload the file again until the message "Done uploading." appears.

You should then see the **red** LED (marked with an "L") on the board start blinking.



Step 9: **Change the delay time to 100**, either HIGH time or LOW time. Upload the Blink sketch again. You should see the time difference in blinking.

****** TA Check 3: Demo to your TA that the blinking time changes according to the different delay times ******

Note: 1. After the Upload button is clicked, the code will be verified first for any syntax error. If no error is found, it will be compiled and uploaded to the Nano-Board.

2. You can do the coding without connecting the Nano-Board to the PC. Simply verify your codes by clicking ✓. When codes are ready, connect the board and do the upload.

Experiment 4: Use Nano-Board to generate PWM signals (~20 min)

Now we use the Nano-board to generate PWM signals for controlling the motor rotation speed.

Step 1: **Download** the "**lab_05_s4**" Arduino sketch program from your Canvas lab page and upload the code to your Nano-board.

```
const int pinL_PWM = 9;          //pin D9
const int pinR_PWM = 11;         //pin D11

void setup()
{
  pinMode(pinL_PWM, OUTPUT);
  pinMode(pinR_PWM, OUTPUT);
}

void loop()
{
  analogWrite(pinL_PWM, 200);
  analogWrite(pinR_PWM, 100);
}
```

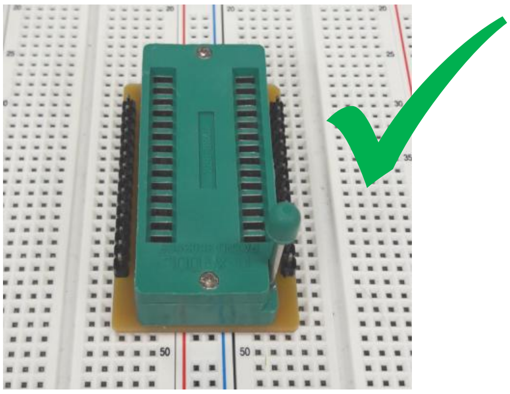
Notice:

- There are two motors on your robot car, you need to generate two PWM signals to control each of the two motors.

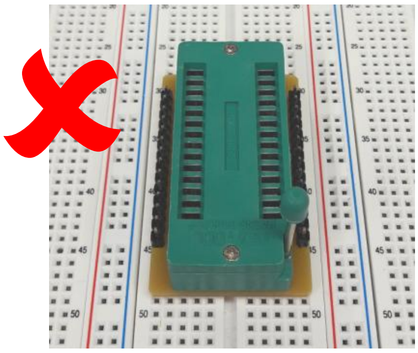
- In the coding text above, we use pin 9 of your Nano-board to generate the PWM signal for your Left motor (pin 9: **L_PWM**) and pin 11 to generate the PWM signal for your Right motor (pin 11: **R_PWM**).

- We use the **"analogWrite"** function to assign the values of the duty cycle to each PWM signal and notice the difference between L_PWM & R_PWM with different values: L_PWM with 200 while R_PWM with 100.

Step 2: Insert the given IC socket into your breadboard. Note that you need to locate it as shown in the figure below.

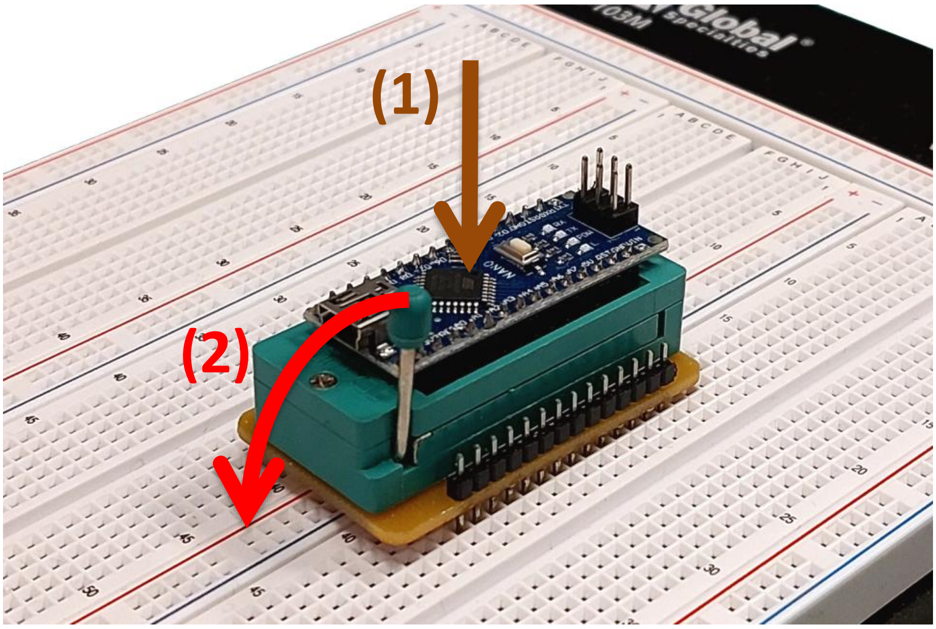


The 2-column breadboard strip is under the IC socket.



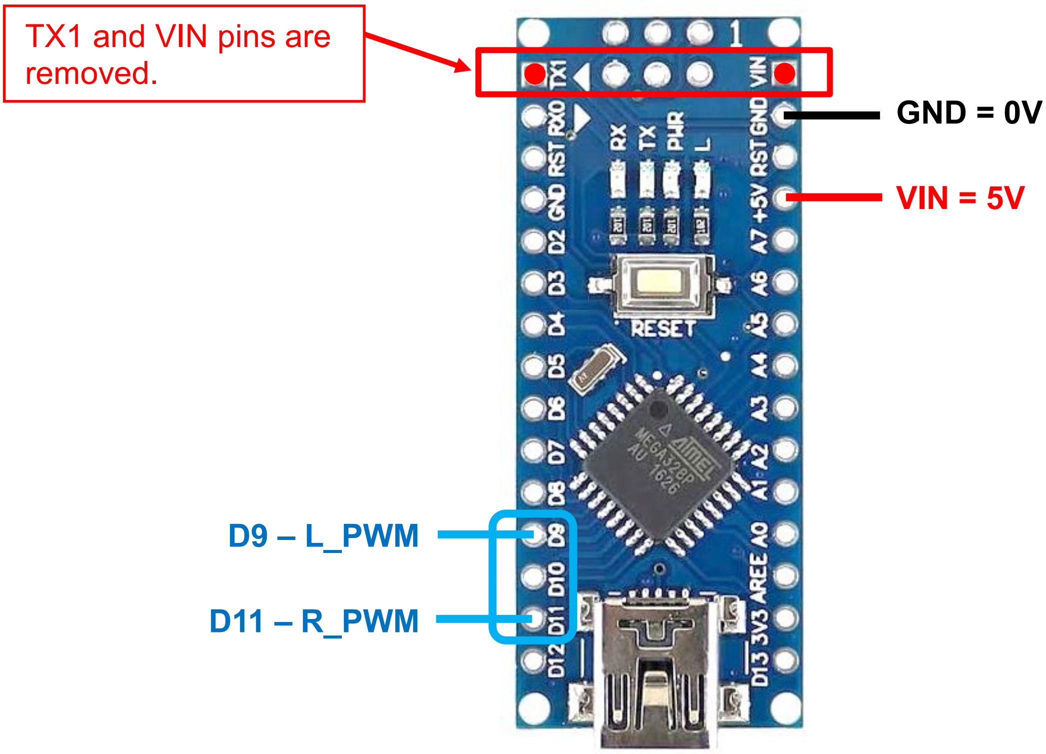
No available holes for wiring out.

Step 3: Disconnect the Nano-Board from the USB cable, and: (1) put it into the IC socket; (2) pull down the handle to lock it.



Note: When taking away the Nano-board from the IC socket, pull up the handle first.

Step 4: Use a DC power supply to generate a 5V source. Use wires to connect your Nano-board to the generated 5V as shown below.



Step 5: Use the DSO to observe the two PWM waveforms generated at D9 & D11. Connect CH1 to measure D9 (L_PWM) while CH2 measuring D11 (R_PWM). You should see the difference between the waveforms.

Q1: Sketch what you see on the DSO (both CH1 & CH2 waves, including the voltage & time values of each division as displayed on the screen).

Q2: What is the frequency (as shown on the screen) of the L_PWM and R_PWM signals?

Q3: What is the duty cycle (give a rough number) of the L_PWM and R_PWM signals?

******TA Check 4: Demo to your TA the generated waveforms from Nano-board ******

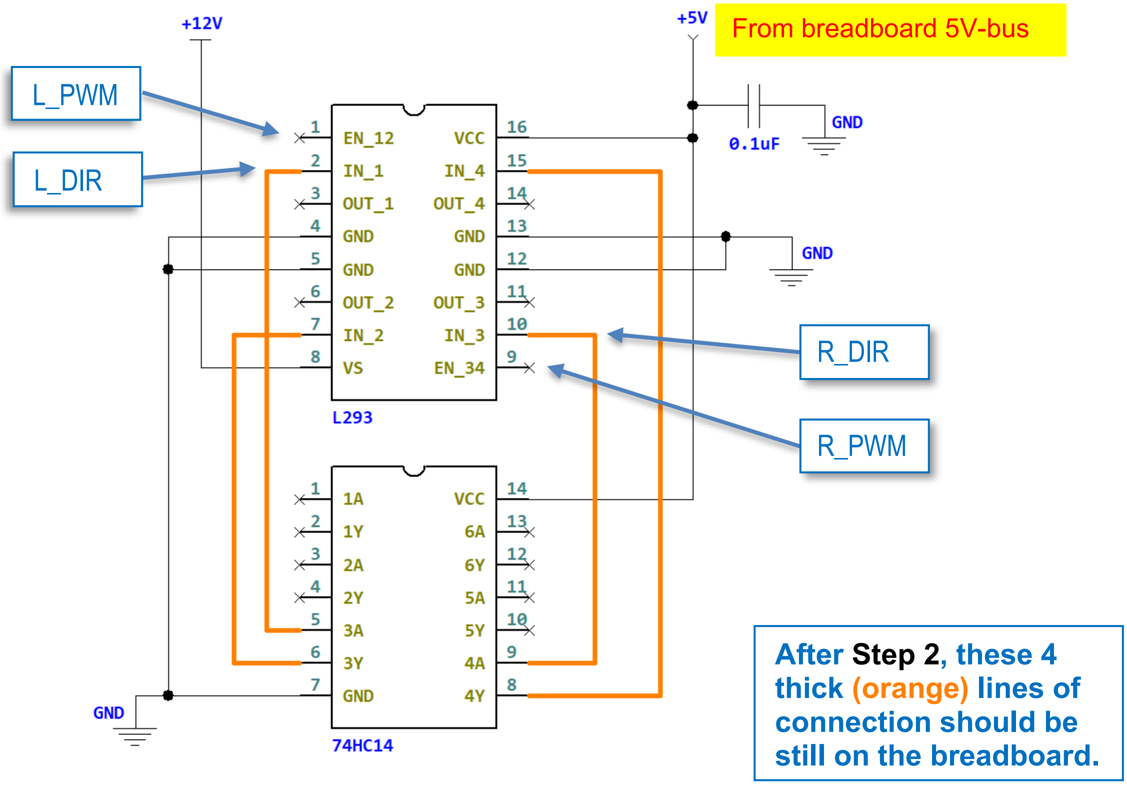
Experiment 5: Use sensors to control motor rotation (~30 min)

In this experiment, we need to use your previous work from **Lab4**.

Find the **L293 H-bridge** circuit on your breadboard and do the below modifications.

Step 1: **Remove** the wires connecting **L_PWM** and **L_DIR** (pins **1** and **2** of your L293 H-bridge) from the **5V** or **GND** bus on your breadboard.

Step 2: **Remove** the wires connecting **R_PWM** and **R_DIR** (pins **9** and **10** of your L293 H-bridge) from the **5V** or **GND** bus on your breadboard.



Step 3: **Rewire** your Nano-board connections to ensure that its **5V** and **GND** are connected to the **5V-bus** or **GND-bus** on your breadboard as you managed in Lab4.

Step 4: Apply 12V to your LM7805 input and measure its output to verify the regulated 5V.

You should only have a 12V generated from the DC power supply. Anywhere a 5V is needed it should be connected to the LM7805 output through your breadboard 5V-bus.

Be clear about where the 12V and 5V (LM7805 input $\mathbf{V_{in}}$ and output $\mathbf{V_{out}}$) are on your breadboard.

Do NOT mix up the 12V and 5V, so you won't burn anything.

****** TA Check 5.1: Demo to your TA that the voltage measurements on your breadboard match with the below table**

LM7805 Regulator		**L293** Motor Driver		**74HC14** Inverter		**Nano-board**
$\mathbf{V_{in}}$	$\mathbf{V_{out}}$	Pin **8**	Pin **16**	Pin **7**	Pin **14**	+5V terminal
12V	**5V**	**12V**	**5V**	**0V**	**5V**	**5V**

****

TURN OFF the 12V power. Now we use the Nano-board outputs to give **L_PWM** & **L_DIR,** and **R_PWM** & **R_DIR** signals to your L293 H-bridge circuit and link the line sensor you used in **Experiment 2** to the Nano-Board to change the DC motor rotation direction.

Step 5: Connect your Nano-Board outputs (**L_PWM** & **L_DIR,** and **R_PWM** & **R_DIR**) to your L293 H-bridge circuit as shown in the "Complete Circuit Diagram" on page 11.

Step 6: Take two sensors to be **L_Sensor** and **R_Sensor**, respectively. Power and ground your sensors to the **5V-bus** or **GND-bus** on your breadboard and connect sensor outputs (from the white "OUT" wire on each sensor) to the **Nano-board A5** and **A3** terminals as shown in the "Complete Circuit Diagram" on page 11.

Step 7: Connect two DC motors to your L293 H-bridge circuit following the same setup as in Lab4 (refer to page 5 of your Lab4 manual). Insert the four terminals of your left motor (through the breadboard row holes) to the left side of L293 (to pins **3, 4, 5** and **6**), and insert the right motor to the right side of L293 (to pins **11, 12, 13** and **14**).

Step 8: **Download** the "**lab_05_s5**" Arduino sketch program from your Canvas lab page.

Step 9: **Take away the Nano-board from the IC socket** and upload the "**lab_05_s5**" sketch to your Nano-board using that USB cable.

DO NOT do uploading when the Nano-board is still in the IC socket.

This may damage the Nano-board if your circuit is working under high voltage.

Step 10: Put the Nano-board back into the IC socket after uploading the sketch.

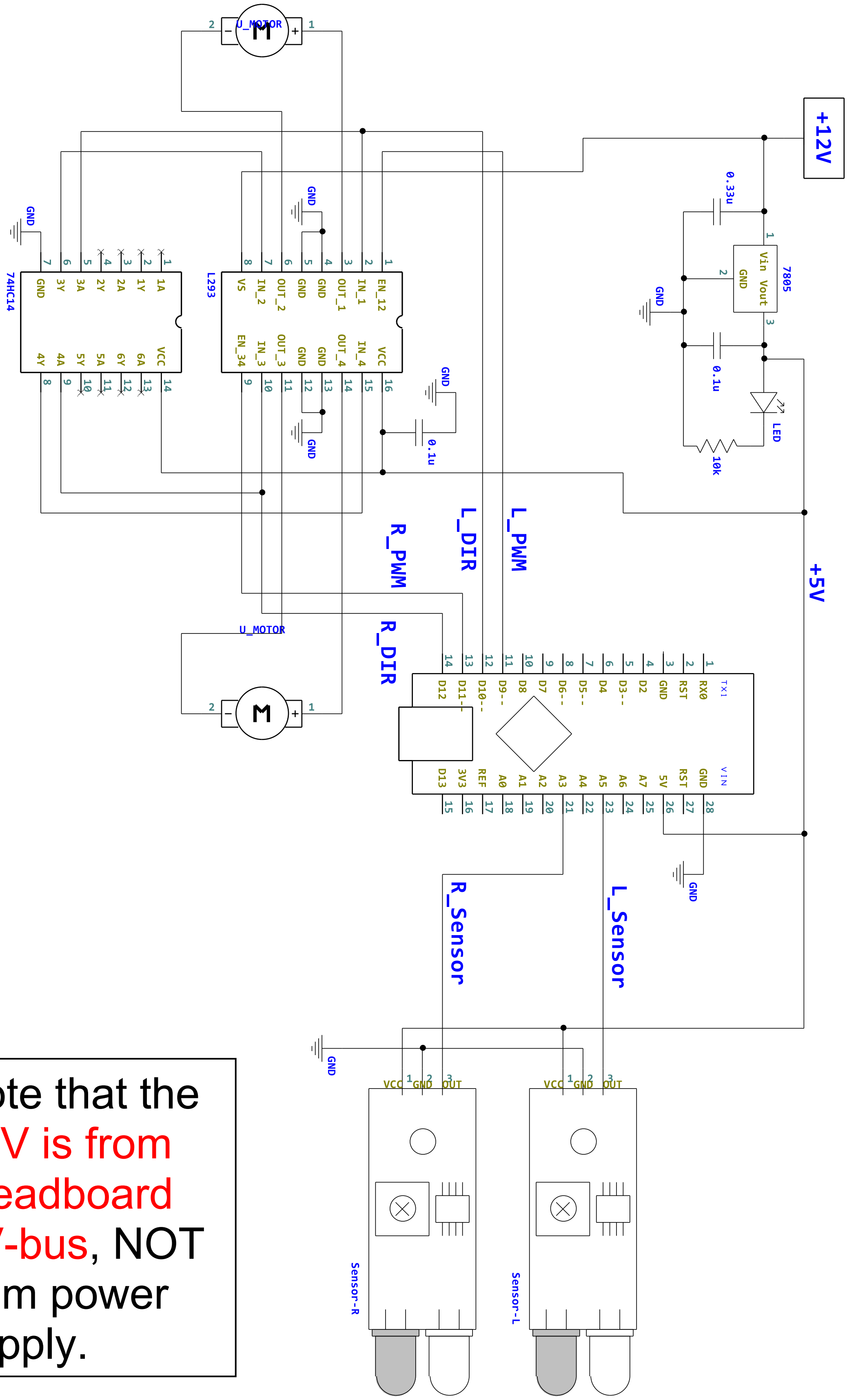
Step 11: Apply 12V to your complete circuit. Have each sensor move across the white line and dark surface.

****** TA Check 5.2: Demo to your TA that the sensor movement can change the rotating direction of each motor ******

KEEP your circuits (LM7805, L293 & 74HC14, and Arduino Nano-board) on the given breadboard as you will use them at future labs!!!

Remember to clean up your bench and **return your project box** and **cable box**! A messy table will cost 1 point!

Appendix: Complete Circuit Diagram for Experiment 5



Note that the +5V is from breadboard 5V-bus, NOT from power supply.